

Measuring Media Bias Toward Reform Prosecutors: A Multi-Method NLP Analysis of Bay Area News Coverage, 2019–2024

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Abstract

Progressive prosecutors have faced sustained media scrutiny, but whether their coverage systematically differs from that of traditional prosecutors, and through what mechanisms, has not been empirically tested. Analyzing 136,313 Bay Area news articles published between 2019 and 2024, I develop a multi-method natural language processing pipeline combining aspect-based sentiment analysis, zero-shot stance classification, keyword theme detection, and media framing analysis to measure how five district attorneys spanning the progressive–traditional spectrum are covered. I find that media bias is not a unitary phenomenon: evaluative framing and emotional tone are empirically independent dimensions. While I detect no systematic difference in emotional tone, progressive prosecutors receive substantially more critical evaluative treatment: an effect that surges during recall campaigns and is confirmed by within-county comparisons. Progressive prosecutors also receive accountability framing at twice the rate of traditional counterparts, who are more often covered through human-interest narratives. These findings demonstrate that the press covers reform prosecutors differently in kind rather than in degree, and that research relying on tone alone systematically misses the evaluative dimension of coverage most consequential for prosecutorial politics and electoral accountability.

1 Introduction

The progressive prosecution movement represents one of the most significant developments in American criminal justice over the past decade. Beginning with the election of reform-minded district attorneys in major cities—including Larry Krasner in Philadelphia (2017), Kim Foxx in Chicago (2016), and Chesa Boudin in San Francisco (2019)—a new generation of prosecutors has sought to reduce mass incarceration, curtail cash bail, decline to prosecute low-level offenses, and hold law enforcement accountable (Bellin, 2019; Sklansky, 2017). These prosecutors have faced intense political opposition, including recall campaigns, legislative challenges, and sustained media scrutiny (Bazon, 2019). Goldstein (2024) characterizes this resistance as “toplash”: elite-driven opposition from wealthy donors, police unions, and political establishments that is distinct from grassroots backlash and specifically targets prosecutors’ reform agendas through media campaigns and institutional pressure.

Media coverage plays a crucial role in shaping public attitudes toward criminal justice reform. How prosecutors are portrayed in the press, whether through sympathetic human-interest narratives or critical accountability frames, can influence public support for reform policies and, ultimately, electoral outcomes (Iyengar, 1991; Entman, 1993). The recall of San Francisco District Attorney Chesa Boudin in June 2022, widely attributed to perceptions of rising crime and prosecutorial leniency, underscores the political salience of media framing in this domain (Yogev, 2026).

Despite the importance of media coverage in the politics of prosecution, there has been little systematic analysis of whether progressive prosecutors receive measurably different treatment from the press compared to their traditional counterparts. Existing scholarship on media bias in criminal justice has focused primarily on race and crime (Dixon and Linz, 2000), sentencing coverage (Surette, 2015), and general ideological slant at the outlet level (Groseclose and Milyo, 2005). The prosecutor-specific question, whether the press covers reform-oriented officeholders more negatively and if so through what mechanisms, remains largely unexamined. The most systematic prior effort, a pilot study of prosecutor coverage across four states, found that the median incumbent

prosecutor appeared in only 24 articles during an entire election year, and 89 percent of that coverage contained no information about prosecutors’ decisions or policies (Hessick and Thornburg, 2023).

This study addresses that gap. I analyze 136,313 news articles from 21 Bay Area publications over a five-year period (2019–2024), comparing coverage of five district attorneys who span the progressive–traditional spectrum. The San Francisco Bay Area provides an unusually strong research design: multiple counties with sequential tenures of progressive and traditional prosecutors in the same jurisdiction allow within-county comparisons that control for local media markets, crime conditions, and editorial cultures.

I contribute to the literature in three ways. First, I develop a multi-method NLP pipeline that goes beyond simple sentiment analysis to measure aspect-based sentiment (tone specifically toward the prosecutor, not the article generally), zero-shot stance classification (whether the article’s framing criticizes or defends the prosecutor), and media frame detection (the narrative lens through which prosecutors are presented). Second, I provide the first large-scale empirical test of differential media treatment of prosecutors by ideological orientation. Third, I demonstrate that framing differences, specifically the disproportionate use of accountability and reform frames for progressive prosecutors, may be more consequential than aggregate tone differences.

2 Literature Review

2.1 Media Framing and Criminal Justice

The study of media framing in criminal justice draws on Entman’s (1993) definition of framing as selecting and emphasizing particular aspects of perceived reality to promote specific interpretations. In criminal justice contexts, framing choices—whether to emphasize individual responsibility versus systemic factors, crime trends versus policy outcomes, victims versus defendants—shape public understanding of complex policy questions (Iyengar, 1991; Gross, 2008). Tversky

and Kahneman (1981) foundational work on framing effects demonstrates that logically equivalent information presented in different frames produces systematically different judgments, a finding repeatedly confirmed in criminal justice settings (Slovic et al., 2000).

Research on crime news specifically has documented a persistent “mean world” effect (Gerbner et al., 2002): local news overrepresents violent crime relative to its actual incidence, contributing to public fear and support for punitive policies. Dixon and Linz (2000) demonstrate that racial minorities are overrepresented as perpetrators and underrepresented as victims in local television news, while Gilliam and Iyengar (2000) show that implicit racial cues in crime coverage activate punitive policy preferences among white viewers. The asymmetry extends to victim portrayal: white victims are significantly more likely than Black victims to receive sympathetic coverage featuring personal photographs and family context (Global Strategy Group and Equal Justice Initiative, 2021). These media-constructed racial associations drive support for punitive policies independently of actual crime rates (Ghandnoosh, 2014), a dynamic directly relevant to reform prosecutors who frame their agendas around addressing racial disparities in prosecution. This body of work establishes that media coverage of criminal justice is neither neutral nor proportional to underlying reality.

These patterns reflect structural features of the media industry rather than individual editorial choices. Beale (2006) argues that commercial pressures determine the news media’s contemporary treatment of crime: sensationalized coverage is cost-effective and audience-attracting, creating persistent overrepresentation of violent crime. Market incentives also generate structural dependencies on law enforcement as readily available, authoritative sources, a reliance that shapes the narrative framework within which prosecutors are evaluated (Beale, 2006; Surette, 2015). These media dynamics have measurable downstream consequences: Ash and Poyker (2024) provide causal evidence that exposure to conservative crime coverage increases sentencing harshness among elected judges, while Romer et al. (2003) confirm that local television news viewing increases fear of crime independent of actual crime rates.

2.2 Prosecutorial Politics and Public Perception

The rise of the progressive prosecution movement has generated a growing scholarly literature on prosecutorial discretion, electoral accountability, and political backlash (Pfaff, 2017; Sklansky, 2017; Bellin, 2019). Wright (2009) provides a framework for understanding prosecutorial elections as mechanisms of democratic accountability, while Davis (2007) documents the broad discretionary powers that make the prosecutor's office a site of significant policy variation. Bazelon (2019) chronicles the political dynamics surrounding progressive prosecutors, noting that opponents have mobilized law enforcement unions, victims' rights organizations, and media campaigns to challenge reform agendas. Police unions occupy a dual position in this dynamic: they serve as primary media sources for crime stories while simultaneously funding opposition campaigns against reform prosecutors and lobbying for legislation limiting prosecutorial discretion (Goldstein, 2024). This structural asymmetry means that the most frequently quoted institutional actors in crime coverage are also the most organized opponents of prosecutorial reform. Recent reviews document the scope of this opposition, including legislative challenges across multiple states and coordinated recall campaigns (Mitchell and Petersen, 2025).

However, empirical analysis of media coverage of individual prosecutors remains scarce. Despite prosecutors' central role in criminal justice, they are structurally under-covered: a UNC pilot study found that prosecutors receive minimal media attention, with coverage rarely mentioning their elected status even during election years (Hessick and Thornburg, 2023). This coverage deficit is stark relative to law enforcement; in the Bay Area corpus analyzed here, 82 percent of crime articles mention police compared to 15 percent that mention the district attorney. The scarce attention that prosecutors do receive is thus particularly high-stakes, as it constitutes the primary channel through which the public evaluates prosecutorial policy. Most studies of media bias in politics focus on candidate coverage in electoral contexts (D'Alessio and Allen, 2000; Puglisi and Snyder, 2011) or ideological slant at the outlet level (Grosseclose and Milyo, 2005; Gentzkow and Shapiro, 2010). The prosecutor-specific question is distinct because prosecutors are both political figures

and quasi-judicial actors, operating in a policy domain, criminal justice, where media coverage has documented and substantial effects on public attitudes.

2.3 Computational Text Analysis in Political Communication

Recent advances in natural language processing have expanded the toolkit available for media bias measurement. Transformer-based models, including BERT (Devlin et al., 2019) and its derivatives, have achieved state-of-the-art performance on sentiment analysis, stance detection, and text classification tasks (Liu et al., 2019). Zero-shot classification approaches, which leverage models trained on natural language inference to classify text without task-specific training data, are particularly valuable in political communication research where labeled training corpora are scarce (Yin et al., 2019).

Applications of NLP to media bias detection include Hamborg et al. (2019), who develop a framework for identifying framing bias through word choice and emphasis, and Fan et al. (2019), who use BERT-based models to detect informational and lexical bias in news articles. Card et al. (2015) apply latent frame analysis to immigration coverage, demonstrating that computational methods can reliably detect framing patterns at scale. This study builds on these approaches by combining multiple NLP methods—sentiment analysis, stance detection, and frame classification—into a composite measure that captures different dimensions of media treatment.

3 Data and Methods

3.1 Data

The analysis draws on a corpus of 136,313 news articles published between January 2019 and December 2024, collected from LexisNexis Academic across 21 Bay Area publications. The corpus includes major regional outlets (*San Francisco Chronicle*, *sfgate.com*, *East Bay Times*, *Mercury News*), television station web content (*kron4.com*, *ktvu.com*, *nbcbayarea.com*, *cbsnews.com*), and

digital-native local news sources (berkeleyside.com, Palo Alto Online). The temporal range spans the full tenure of Chesa Boudin as San Francisco District Attorney (2020–2022), the subsequent tenure of his replacement Brooke Jenkins (2022–present), the transition from Nancy O’Malley to Pamela Price in Alameda County (2023), and the ongoing tenure of Steve Wagstaffe in San Mateo County.

3.2 Article Filtering and Prosecutor Attribution

Not all articles in the corpus concern crime and criminal justice. I apply a two-stage relevance filter: first, a keyword-based pre-filter requiring at least two crime/justice terms (prosecutor, arrest, sentencing, felony, etc.), which retains 107,713 articles; second, for future refinement, a zero-shot classifier (BART-MNLI; [Lewis et al. 2020](#)) to distinguish crime/justice articles from other topics. Prosecutor attribution uses regex-based name matching across known variants (e.g., “Boudin,” “Chesa Boudin,” “DA Boudin”) with special disambiguation logic for “Price” (counted as Pamela Price only within 50 words of contextual terms such as “DA,” “prosecutor,” “Alameda,” or “Pamela”). Articles are assigned a primary prosecutor based on mention frequency, with headline mentions weighted at $3\times$. A disambiguation step ensures that generic last-name matches (e.g., “Jenkins” appearing in articles before Brooke Jenkins took office in July 2022; “Price” in articles about price gouging before Pamela Price took office in January 2023) are validated against first-name co-occurrence, preventing false positive attributions from other individuals sharing the surname or from the common English word “price.” This process yields 12,953 articles attributed to five prosecutors (Table 1). Importantly, the relevance filter uses generic crime and justice vocabulary (“prosecutor,” “arrest,” “felony”) that is orthogonal to the outcome-specific theme dictionaries applied later (“soft-on-crime,” “recall campaign,” “public-safety-failure”); the filtering step therefore does not select articles on the basis of the thematic variables subsequently measured.

Table 1: Prosecutor sample characteristics.

Prosecutor	County	Ideology	Tenure	<i>n</i>
Chesa Boudin	San Francisco	Progressive	2020–2022	5,759
Brooke Jenkins	San Francisco	Traditional	2022–present	3,200
Nancy O’Malley	Alameda	Traditional	2009–2023	1,478
Pamela Price	Alameda	Progressive	2023–2024	828
Steve Wagstaffe	San Mateo	Traditional	2010–present	1,648
<i>Progressive total</i>				6,601
<i>Traditional total</i>				6,352

3.3 Multi-Method Bias Detection

I employ four complementary methods to measure the tone of coverage directed at each prosecutor, combined into an aggregate tone-evaluation index (hereafter “composite”) normalized to $[-1, +1]$ (negative = more negative toward the prosecutor):

Method A: Aspect-based sentiment analysis (weight: 0.35). For each prosecutor mention in an article, I extract a three-sentence context window centered on the mention and compute sentiment polarity using a fine-tuned RoBERTa model (cardiffnlp/twitter-roberta-base-sentiment-latest; Loureiro et al. 2022). The article-level score averages across all mention windows. This approach measures sentiment *toward the prosecutor specifically*, rather than the overall tone of the article, a critical distinction in crime reporting, which is inherently negative in subject matter.

Method B: Zero-shot stance classification (weight: 0.30). For paragraphs containing prosecutor mentions, I classify the text’s stance toward the prosecutor using BART-MNLI (Lewis et al., 2020) with three candidate labels: “This text criticizes the prosecutor’s handling of crime,” “This text defends the prosecutor’s approach,” and “This text is neutral about the prosecutor.” The article-level score is the net balance of critical versus defensive paragraphs.

Method C: Enhanced keyword analysis (weight: 0.20). I apply curated theme dictionaries—including categories for crime-rising, soft-on-crime, recall, case dismissal, releasing criminals, public safety failure, victim neglect, police conflict, and office dysfunction—within three-sentence

windows around prosecutor mentions. A negation detection module (checking three words before each keyword for negation terms) prevents false positives. This method operationalizes the anti-prosecutor framing patterns identified in prior qualitative work.

Method D: Document-level sentiment baseline (weight: 0.15). Whole-article sentiment serves as a control measure, capturing the general negativity of crime coverage independent of prosecutor-specific framing.

Prosecutor-attributed theme detection. To complement the keyword-based theme analysis (Method C), I implement a multi-method theme attribution algorithm that requires critical themes to be *explicitly linked to prosecutors* through co-occurring linguistic patterns rather than simple keyword proximity. The algorithm employs four detection methods: (1) context-aware dictionary matching, in which compound regex patterns require both a prosecutor reference and a theme keyword within the same expression (e.g., “crime rising because of the DA” rather than merely “crime” near “prosecutor”); (2) relationship-based regex patterns capturing causal structures linking prosecutors to outcomes; (3) targeted criticism detection using high-confidence blame and failure patterns; and (4) sentence-level co-occurrence analysis confirming that prosecutor references and criticism keywords appear in the same sentence. Each method produces a score on a 0–25 scale, combined with weights of 0.30, 0.35, 0.15, and 0.20 respectively. A multi-method validation step caps single-method scores at 15, applies a $1.1\times$ multiplier when two methods agree, and a $1.25\times$ multiplier for three or more, ensuring that high scores reflect convergent evidence. Nine themes are tracked: crime rising, releasing criminals, victim neglect, police conflict, office dysfunction, soft on crime, case dismissal, recall, and public safety failure.

The composite score weights reflect a deliberate precision–generality hierarchy: the two prosecutor-targeted measures (aspect-based sentiment, 0.35; stance, 0.30) are prioritized over the two more general text-level measures (keywords, 0.20; document sentiment, 0.15), on the theory that prosecutor-specific precision outweighs broader lexical coverage. This weighting scheme is empirically conservative: it assigns the highest weight (0.35) to the method with the weakest effect (aspect-based

sentiment, $d = 0.037$) and correspondingly lower weights to the methods with larger effects (stance $d = -0.341$ at 0.30; keywords $d = -0.218$ at 0.20), mechanically attenuating the composite rather than inflating it. To verify that results do not depend on this choice, I recomputed the composite using z-standardized method scores under six alternative specifications (equal weights, inverse weights, effect-proportional weights, and three leave-one-out variants; $n = 10,000$ articles with complete scores for all four methods). Across all specifications the composite remains negative and highly significant ($p < 10^{-4}$), with effect sizes ranging from $d = -0.08$ (dropping stance, the strongest single method) to $d = -0.39$ (retaining only the two evaluative methods). The current weighting produces one of the smallest composite effects among all schemes tested.

3.4 Media Framing Analysis

Beyond tone, I analyze *how* prosecutors are framed by classifying prosecutor-containing paragraphs into five media frames using BART-MNLI zero-shot classification:

1. **Accountability:** The prosecutor is held responsible for outcomes (e.g., crime rates, case decisions).
2. **Conflict:** Coverage emphasizes disagreement (prosecutor vs. police, victims, or community).
3. **Consequences:** Focus on policy outcomes and their effects.
4. **Human interest:** Individual stories of victims or defendants as narrative vehicles.
5. **Reform/ideology:** The prosecutor is characterized through an ideological or reformist lens.

Although Method B (stance) and the framing analysis both employ BART-MNLI, they are not measuring the same construct: stance detection classifies whether a paragraph criticizes, defends, or is neutral toward the prosecutor, while framing classification identifies the narrative lens (accountability, conflict, human interest, etc.). To verify empirical independence, I computed pairwise correlations between article-level stance scores and all five frame probability scores ($n = 10,000$). Correlations range from $r = -0.10$ (human interest) to $r = -0.21$ (accountability), far below the

$r > 0.8$ threshold for methodological redundancy. The modest negative values are theoretically expected (more critical stance co-occurs with more accountability framing) and confirm convergent validity rather than collinearity.

3.5 LLM-Based Structural Content Extraction (Appendices A–B)

The four NLP methods above, and the framing analysis, all operate on the *aggregate statistical signal* of text: they assign continuous scores by detecting linguistic patterns across many tokens. This is powerful for measuring the overall tone or evaluative stance of coverage but does not reveal *what specific content* articles contain: who is quoted, what claims are made, and with what evidence. To examine these structural features, I employ a complementary approach using Google’s **langextract** library (Gemini 2.5 Flash), reported fully in Appendices [A](#) and [B](#).

Langextract differs from the NLP methods in three fundamental ways. First, it is *extractive* rather than *classificatory*: instead of assigning a score to an article, it identifies and categorizes discrete content elements—specific source quotes, causal claims, loaded phrases—and anchors each extraction to the exact text span it originates from. This source grounding allows for verification and provides a direct link between extracted facts and the article text. Second, it operates with *explicit structured schemas*: the analyst specifies categories of interest in advance (source role, claim type, evidence quality, bias mechanism), and the model returns instances that fit those categories, rather than learning latent patterns from a training corpus. Third, because the extraction schema is analyst-defined and the model is prompted with diverse few-shot examples, the approach is more interpretable and auditable than black-box transformer classifiers, but it is also more dependent on schema design choices and may miss patterns not anticipated by the analyst.

The conceptual distinction matters for interpreting results. The NLP methods measure the *how* of coverage (how critical, how emotionally charged, how evaluatively framed); langextract measures the *what* (what sources appear, what specific claims are asserted, what journalistic standards are violated). The two approaches are therefore complementary rather than redundant. When both

converge, as in the excess of prosecutor-attributed harm claims (Appendix A) alongside the excess of loaded negative language (Appendix B) in progressive prosecutor coverage, the evidence is stronger than either approach alone. When they diverge, as when the overall bias indicator result is null (Appendix B) despite significant stance and keyword effects (Section 4), the divergence reveals that the differential operates through aggregate framing mechanisms rather than discrete, identifiable journalistic violations.

3.6 Statistical Approach

I employ six analytical strategies, applied both to a weighted composite and to each method individually: (1) group-level comparison (Welch’s t -test, Mann–Whitney U , Cohen’s d , Cliff’s delta, bootstrap 95% CIs); (2) same-county paired comparisons (Boudin vs. Jenkins in San Francisco; Price vs. O’Malley in Alameda); (3) OLS regression of bias score on prosecutor ideology, controlling for county, year, and article length, with cluster-robust standard errors by publication; (4) framing differential analysis (chi-square tests on dominant frame frequencies by prosecutor type); (5) interrupted time series and quarterly temporal heterogeneity analysis comparing coverage before and after prosecutor transitions; and (6) TOST equivalence testing (Lakens, 2017) with an equivalence bound of $d = 0.2$. The $d = 0.2$ bound corresponds to Cohen’s definition of a “small” effect and is standard in equivalence testing applications. Crucially, in a large- N study ($n > 13,000$ articles) trivially small effects achieve statistical significance, so the TOST serves as a meaningful practical check: an equivalence result ($p = .007$) indicates the composite effect is not only small by conventional standards but falls within a range generally considered substantively negligible. The per-method analyses are critical: by running each test separately for stance, sentiment, keywords, and document tone, I can identify *which dimensions* of coverage differ by ideology rather than relying on a composite that may average opposing signals to zero. Throughout, p -values below .001 are reported as $p < .001$; exponential notation (e.g., $p < 10^{-127}$) is used selectively where the extreme magnitude is itself informative.

4 Results

4.1 Bias Is Not One Thing: Per-Method Decomposition

The central finding of this study is that “media bias” toward prosecutors is not a single phenomenon. The four bias detection methods, each measuring a distinct dimension of coverage, produce sharply divergent results (Figure 1):

- **Stance classification (B)** produces the largest effect ($d = -0.341$, $p < .001$): articles about progressive prosecutors contain significantly more critical and fewer defensive paragraphs. This method captures evaluative framing: whether coverage *judges* the prosecutor favorably or unfavorably.
- **Keyword analysis (C)** shows a small-to-moderate effect ($d = -0.218$, $p < .001$): anti-prosecutor themes (soft-on-crime, crime-rising, recall) appear more frequently near mentions of progressive prosecutors. This method captures thematic emphasis.
- **Aspect-based sentiment (A)** shows a negligible effect ($d = 0.037$, $p = .058$): the emotional tone surrounding prosecutor mentions is essentially equivalent across groups. This method captures emotional valence.
- **Document-level sentiment (D)** shows a small reversed effect ($d = 0.047$, $p = .008$): articles about traditional prosecutors are slightly *more* negative in overall tone, reflecting the inherent negativity of crime reporting rather than prosecutor-specific bias.

The divergence between stance ($d = -0.341$) and sentiment ($d = 0.037$) is the study’s key conceptual finding: media coverage of progressive prosecutors maintains a neutral emotional tone while adopting a critical evaluative stance. The press covers progressive prosecutors in the same emotional register as their traditional counterparts but evaluates their performance more critically. Figure 2 visualizes this tone–evaluation divergence at the article level.

This pattern has a direct methodological implication: studies relying solely on sentiment analysis would conclude that media treatment of prosecutors is equivalent. The differential emerges only

when measurement captures evaluative framing, a dimension invisible to emotional tone detection.

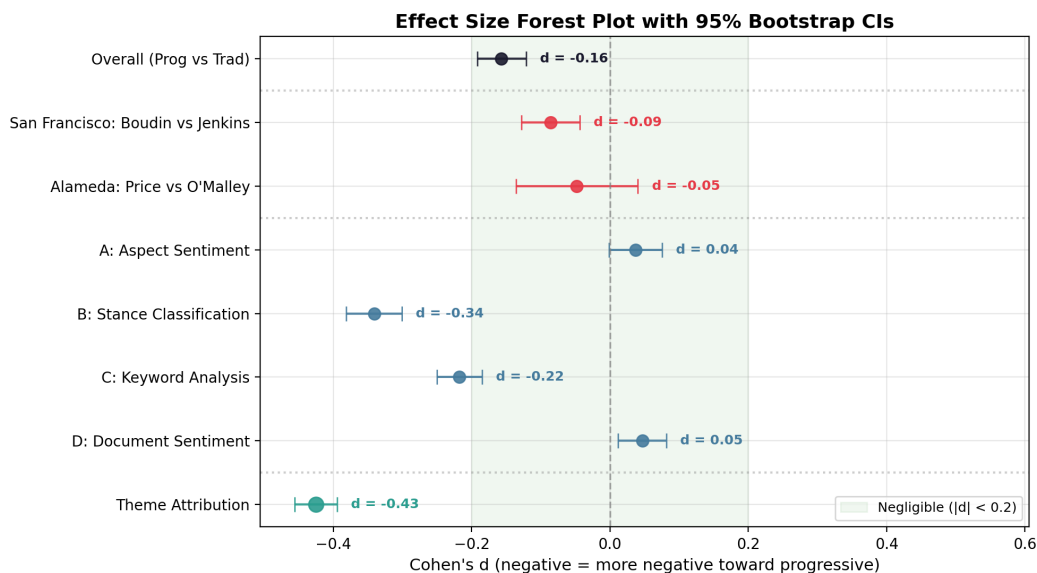


Figure 1: Effect size forest plot with 95% bootstrap confidence intervals. The green band marks the negligible effect zone ($|d| < 0.2$). Results are grouped by analysis type: overall, paired-county, per-method, and theme attribution. Theme attribution ($d = -0.43$) is the study's largest effect.

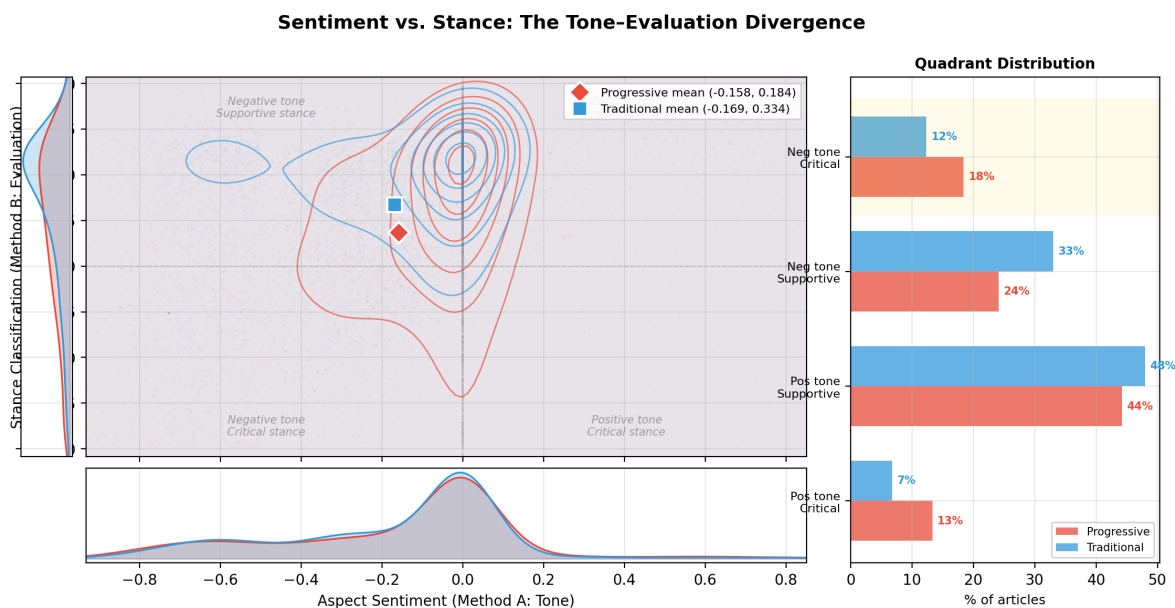


Figure 2: Tone-evaluation divergence at the article level. Central panel: KDE contour plot of Method A (aspect sentiment) vs. Method B (stance classification) scores for progressive (red) and traditional (blue) prosecutors. The horizontal separation in stance contrasts with the overlapping sentiment distributions, confirming that media coverage differs in evaluative framing rather than emotional tone.

4.2 Same-County Paired Comparisons by Method

The within-county design provides the strongest quasi-experimental identification. Per-method analysis reveals *which dimensions* of coverage differ within each jurisdiction:

San Francisco (Boudin vs. Jenkins). Stance classification detects significant differential treatment ($d = -0.207$, $p < .001$) and keyword analysis confirms it ($d = -0.116$, $p < .001$). Aspect sentiment shows no significant difference ($d = 0.033$, $p = .180$), and document sentiment is reversed ($d = 0.058$, $p = .009$). The bias is evaluative, not tonal.

Alameda County (Price vs. O'Malley). On a composite measure, this pair shows a non-significant difference ($d = -0.048$, $p = .295$), an apparent null. However, decomposition by method reveals that the null is an artifact of aggregation: stance classification ($d = -0.275$, $p < .001$) and keyword analysis ($d = -0.373$, $p < .001$) both detect substantial differential treatment, while sentiment ($d = 0.082$, $p = .104$) and document tone ($d = 0.124$, $p = .005$) pull in the *opposite* direction. The composite averages these opposing signals into a near-zero result, masking real differential treatment that operates through evaluative framing.

Table 2: Per-method Cohen’s d for same-county paired comparisons. Significant results ($p < .05$) in bold.

Method	San Francisco		Alameda	
	d	p	d	p
B: Stance	−0.207	< .001	−0.275	< .001
C: Keywords	−0.116	< .001	−0.373	< .001
A: Aspect sentiment	0.033	.180	0.082	.104
D: Document sentiment	0.058	.009	0.124	.005
Composite	−0.086	< .001	−0.048	.295

Figure 3 displays per-prosecutor mean composite scores with 95% confidence intervals. Figure 4 presents the paired county comparisons.

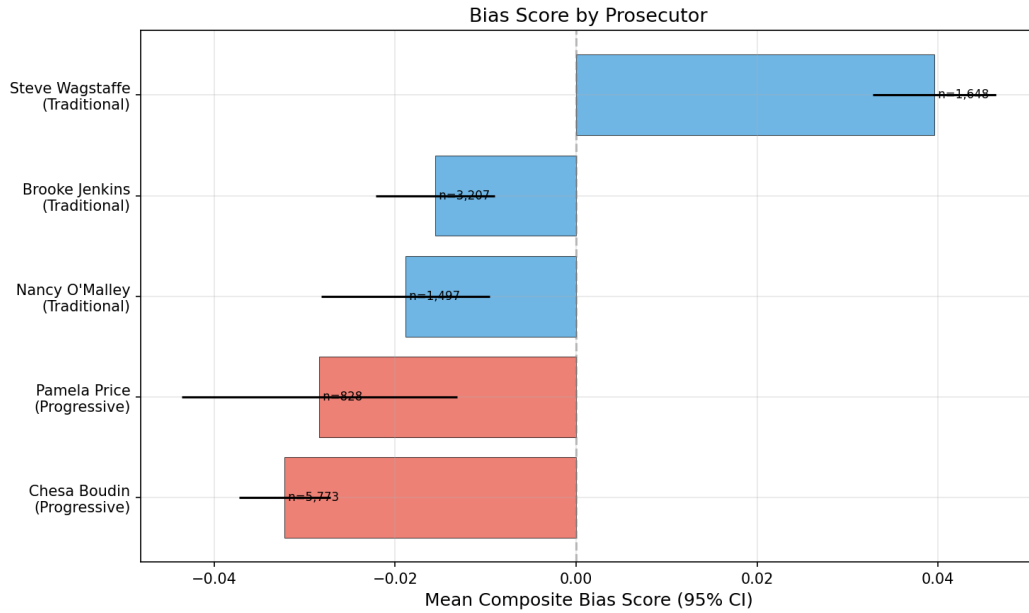


Figure 3: Mean aggregate tone-evaluation index by prosecutor with 95% confidence intervals. Red bars indicate progressive prosecutors; blue bars indicate traditional prosecutors.

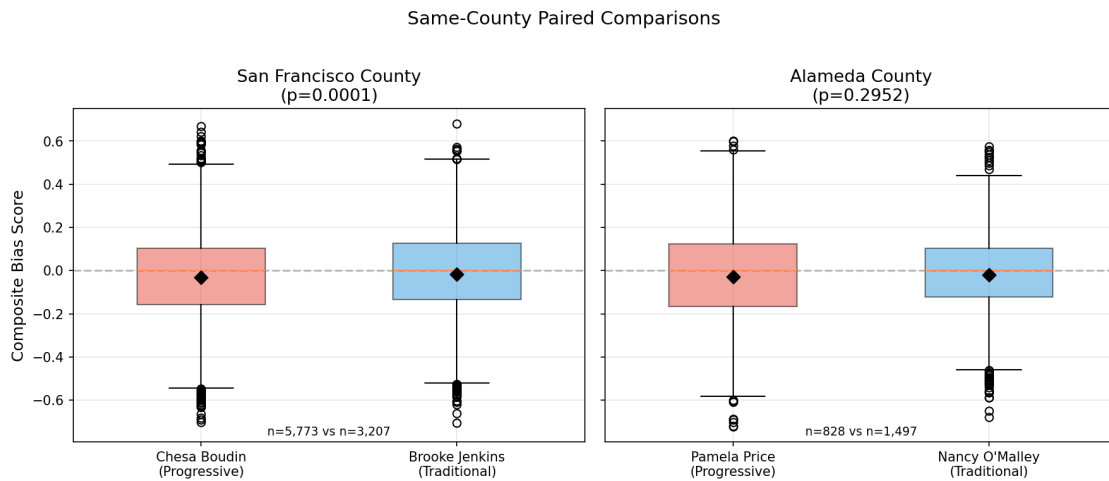


Figure 4: Same-county paired comparisons: San Francisco (Boudin vs. Jenkins) and Alameda (Price vs. O'Malley). Diamond markers indicate group means.

4.3 Regression Analysis by Method

To test whether each bias dimension survives controls for county, year, article length, and publication clustering, I estimate separate OLS regressions for each method (Figure 5):

Table 3: Per-method OLS regressions: progressive prosecutor coefficient with cluster-robust SEs.

Method	β	SE	p	95% CI	R^2
B: Stance	− 0.132	0.030	< .001	[−0.190, −0.074]	.071
C: Keywords	− 0.011	0.002	< .001	[−0.015, −0.007]	.053
A: Aspect sentiment	0.008	0.016	.618	[−0.024, 0.040]	.008
D: Document sentiment	0.017	0.014	.219	[−0.010, 0.044]	.006
Composite	−0.022	0.013	.089	[−0.047, 0.003]	.022

Stance classification and keyword analysis, the evaluative and thematic dimensions, both show significant progressive coefficients that survive all controls. Aspect sentiment and document sentiment do not. The composite regression falls to non-significance ($p = .089$) precisely because it averages the strong evaluative signal with the null tonal signal. Running regressions per method resolves this: evaluative bias is robust to controls; tonal bias does not exist.

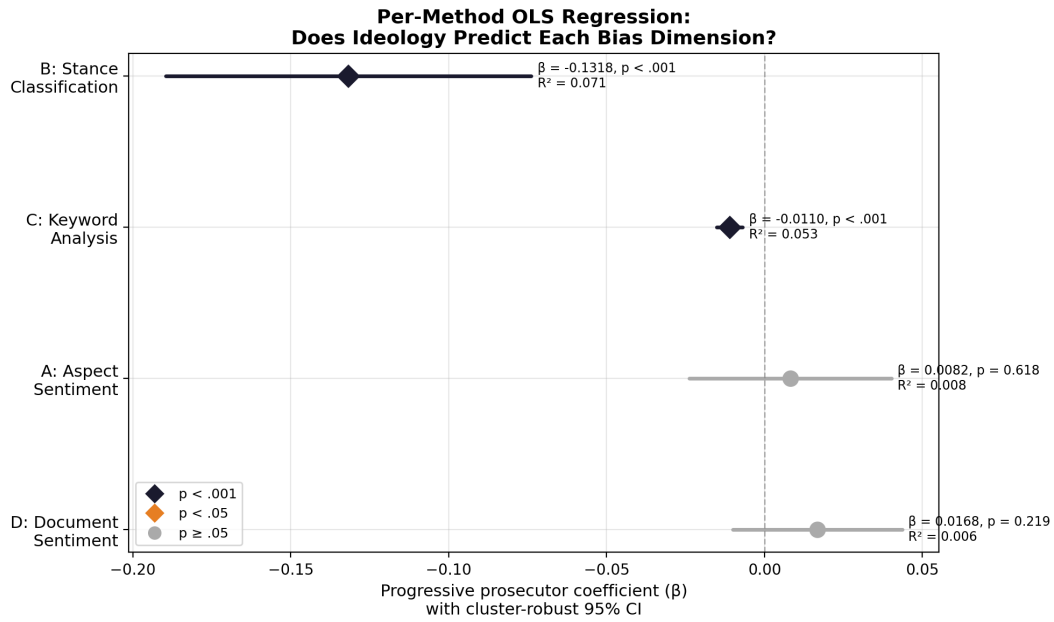


Figure 5: Per-method OLS regression: progressive prosecutor coefficient with 95% cluster-robust CIs. Stance and keyword methods survive controls; sentiment methods do not.

4.4 Framing Analysis

The distribution of dominant media frames differs significantly between progressive and traditional prosecutors, $\chi^2(4) = 443.75$, $p < .001$, Cramér’s $V = 0.225$ (see Figure 6):

Table 4: Dominant media frame distribution by prosecutor ideology.

Frame	Progressive	Traditional
Accountability	39.7%	22.9%
Conflict	30.9%	36.5%
Human interest	21.4%	35.6%
Reform	4.3%	1.3%
Consequences	3.7%	3.7%

Progressive prosecutors are nearly twice as likely to be covered through an accountability frame (39.7% vs. 22.9%, $d = 0.37$, $p < .001$) and more than three times more likely through a reform frame (4.3% vs. 1.3%, $d = 0.19$, $p < .001$). Conversely, traditional prosecutors receive substantially more human-interest framing (35.6% vs. 21.4%, $d = -0.32$, $p < .001$): stories organized around individual victims, defendants, or community members rather than evaluations of prosecutorial performance. Traditional prosecutors also receive slightly more conflict framing (36.5% vs. 30.9%, $d = -0.12$, $p < .001$), while consequences framing is equally distributed ($d = -0.01$, $p = .713$). Effect sizes here reflect per-frame dominant-frame assignment rates, matching the proportions in Table 4.

These framing patterns are consistent with the stance classification finding: the media evaluates progressive prosecutors through accountability and ideological lenses while presenting traditional prosecutors through less evaluative, story-driven narratives.

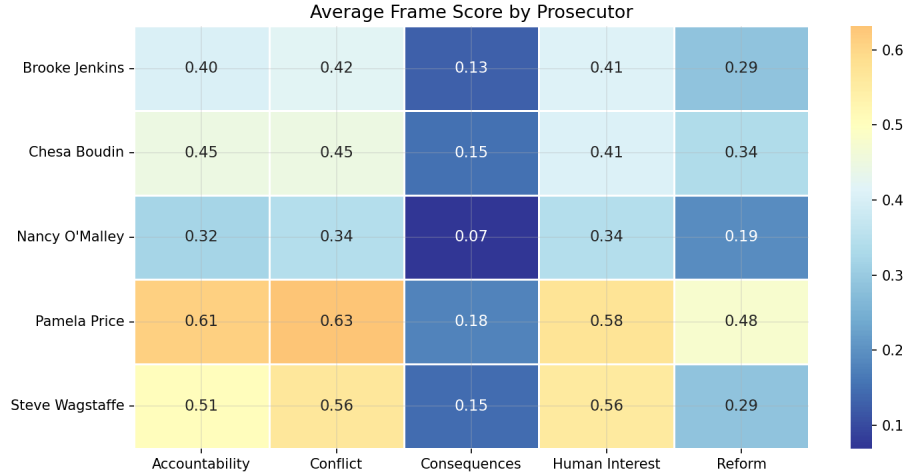


Figure 6: Heatmap of average frame scores by prosecutor. Warmer colors indicate higher frame prevalence. Progressive prosecutors (Boudin, Price) score higher on accountability and reform; traditional prosecutors (Jenkins, O’Malley, Wagstaffe) score higher on human interest. Note: values represent average model probability scores, whereas Table 4 reports dominant frame assignments (the frame with the highest probability per paragraph).

4.5 Temporal Dynamics by Method

Per-method temporal analysis provides the most direct test of whether the evaluative bias identified above is stable or event-driven (Figure 7).

Interrupted time series. At the San Francisco transition (Boudin to Jenkins, July 2022), no method shows a substantively significant shift: stance ($d = -0.049$, $p = .042$), keywords ($d = 0.036$, $p = .092$), sentiment ($d = 0.045$, $p = .058$), and document ($d = 0.049$, $p = .024$) all produce negligible effects. The general negativity of SF DA coverage persists regardless of who holds office. At the Alameda transition (O’Malley to Price, January 2023), however, the per-method pattern is stark: stance ($d = 0.431$, $p < .001$) and keywords ($d = 0.357$, $p < .001$) show large shifts toward more critical coverage of Price, while sentiment ($d = 0.036$, $p = .468$) and document tone ($d = -0.032$, $p = .456$) show no change whatsoever. The evaluative dimensions respond to the prosecutor transition; the tonal dimensions do not.

Quarterly heterogeneity. Figure 7 decomposes the temporal patterns of each method across 21

quarters. Stance classification ($SD = 0.32$, range $[-0.66, +0.30]$) and keyword analysis ($SD = 0.35$, range $[-0.94, +0.44]$) show dramatic quarterly variation, with the deepest effects coinciding with the Boudin recall (2021Q4–2022Q2) and the Price recall movement (2023Q2–Q3). Aspect sentiment ($SD = 0.28$, range $[-0.30, +0.75]$) and document sentiment ($SD = 0.22$, range $[-0.15, +0.69]$) show less variation and no systematic pattern tied to political events. The monthly time series (Figure 8) overlays a coverage-volume-weighted three-month rolling average alongside the raw monthly means; the weighted and unweighted trends are nearly identical, confirming that the temporal patterns are not artifacts of high-variance low-coverage months.

This per-method temporal decomposition resolves a key ambiguity in the aggregate analysis. The composite quarterly d (range $[-0.39, +0.84]$, $SD = 0.37$) appears to show dramatic cyclical variation, but this aggregate masks the distinct temporal signatures of its component methods. Evaluative bias (stance, keywords) is event-driven, surging during recall campaigns and subsiding between them. Tonal bias (sentiment) is absent throughout. The aggregate temporal variation is driven entirely by the evaluative dimensions.

Per-Method Temporal Heterogeneity: Quarterly Effect Sizes

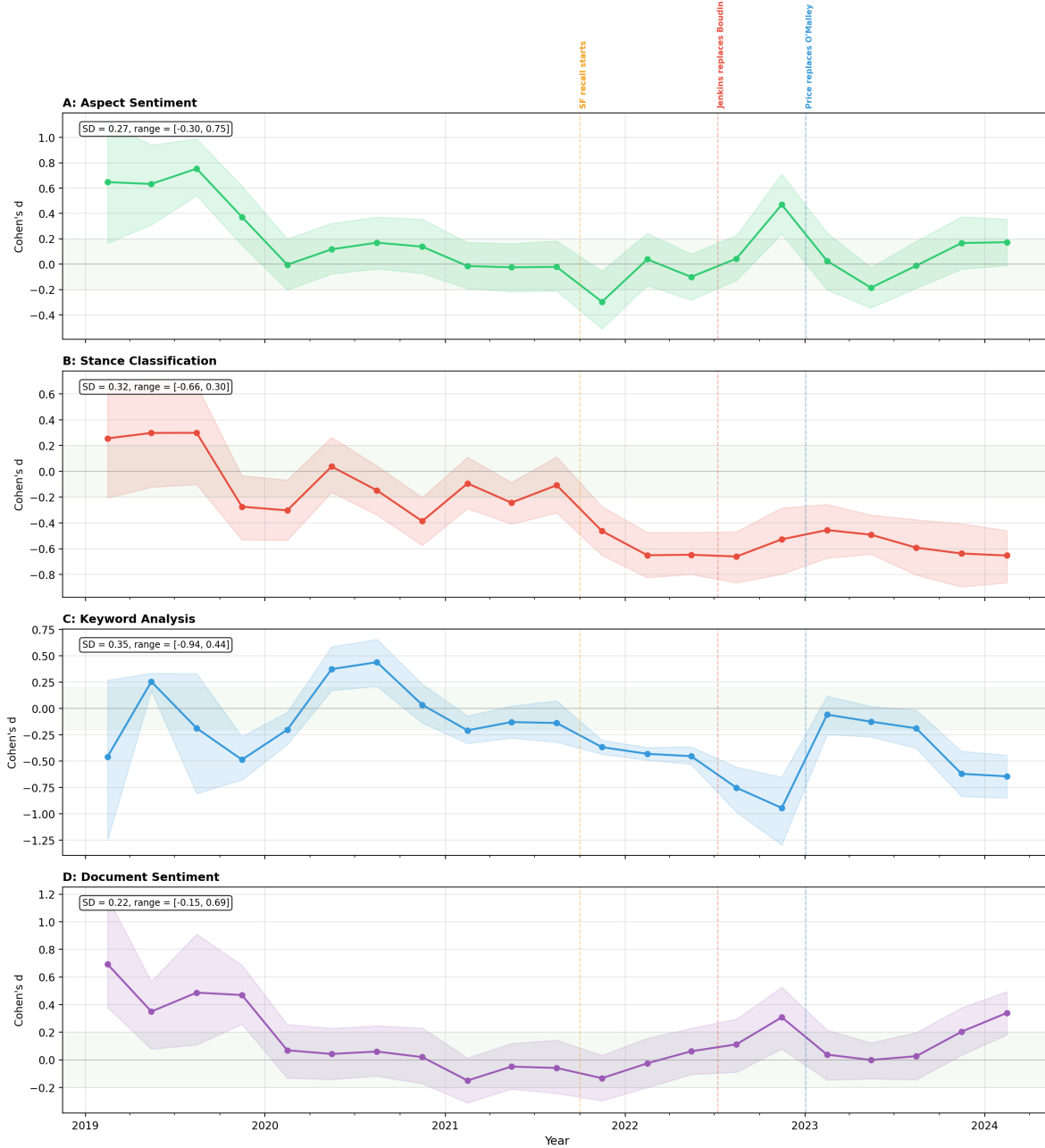


Figure 7: Per-method quarterly Cohen's d (progressive vs. traditional) with 95% bootstrap CIs. The green band marks the negligible zone ($|d| < 0.2$). Dashed lines mark prosecutor transitions. Stance and keyword methods show event-driven variation; sentiment methods remain flat throughout.

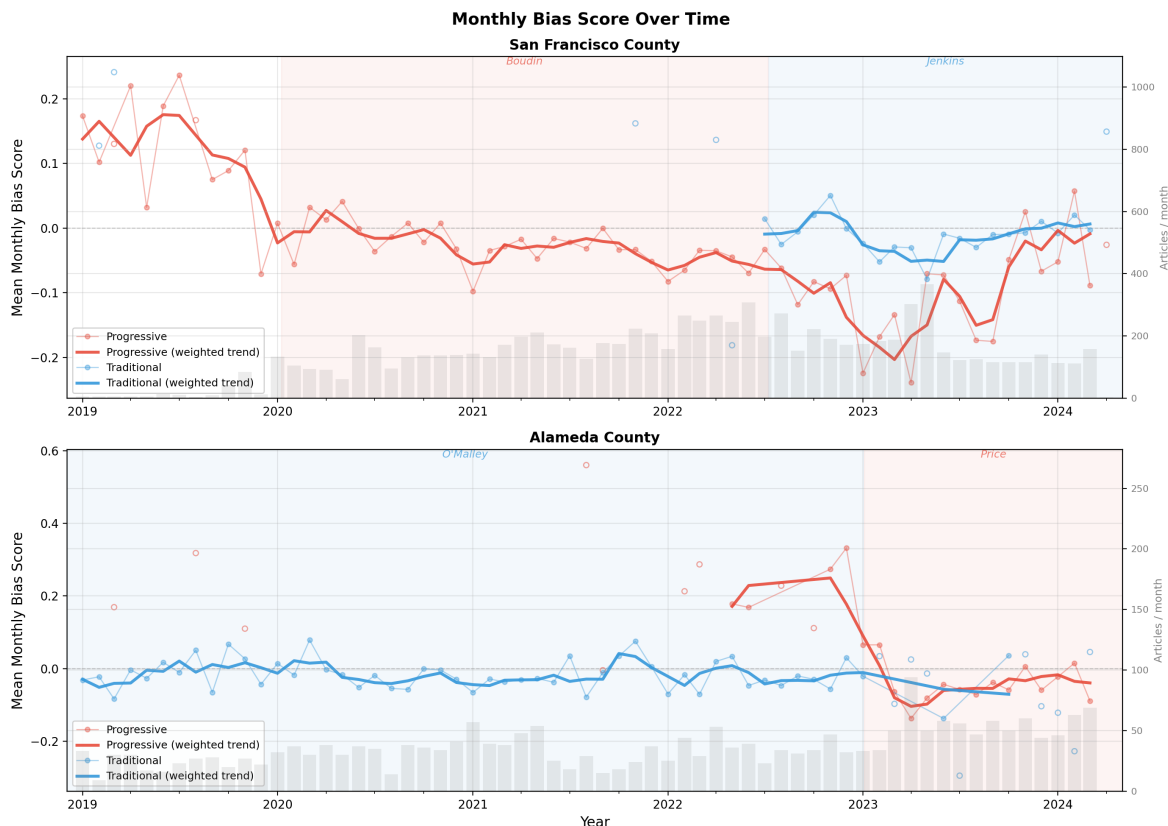


Figure 8: Monthly mean composite scores for San Francisco (top) and Alameda County (bottom), 2019–2024. Faint dots and lines show raw monthly means; thick lines show coverage-volume-weighted three-month rolling averages. Open circles mark months with fewer than five articles. Shaded bands indicate prosecutor tenures; red = progressive, blue = traditional.

4.6 Theme Prevalence and Prosecutor-Attributed Themes

Keyword theme analysis (Method C; Figure 9) measures term density within three-sentence windows around prosecutor mentions. By this measure, recall-related language dominates progressive prosecutor coverage (9.8% vs. 2.7% for traditional), driven overwhelmingly by Boudin’s 2022 recall campaign. After recall, the most prevalent anti-prosecutor themes for progressive coverage are soft-on-crime (1.6% vs. 0.7%), crime-rising (1.5% vs. 0.3%), and public-safety-failure (1.2% vs. 0.3%). The “releasing criminals” theme is the only category where traditional prosecutors receive higher prevalence (5.5% vs. 4.1%), likely reflecting routine bail and parole coverage rather than ideologically targeted framing.

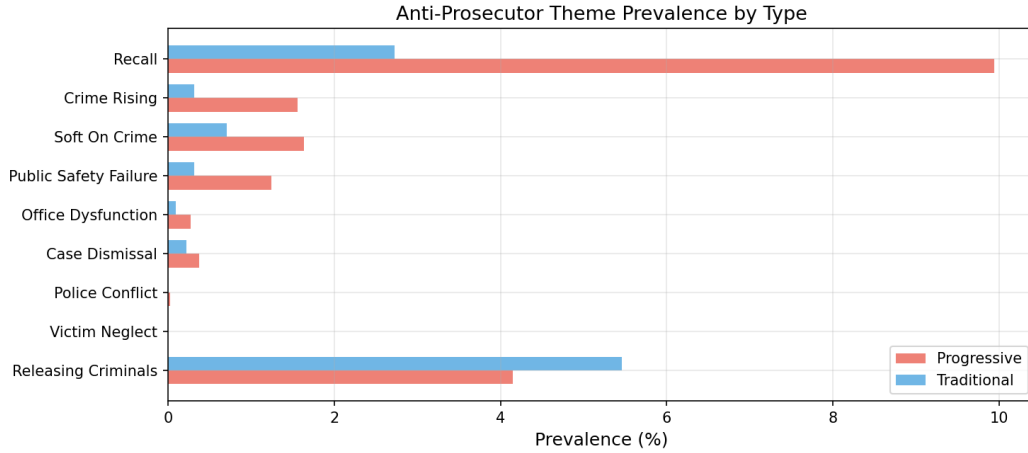


Figure 9: Anti-prosecutor theme prevalence by ideology. Themes are sorted by the difference between progressive and traditional prevalence rates.

The multi-method theme attribution analysis, which requires themes to be explicitly linked to prosecutors through compound regex patterns rather than simple keyword proximity, provides the strongest evidence of differential thematic coverage. Progressive prosecutors receive significantly higher theme attribution scores ($M = 1.80$, $SD = 3.07$) than traditional prosecutors ($M = 0.74$, $SD = 1.71$), $t(10,765) = 24.43$, $p < 10^{-127}$, $d = -0.43$ (signed to match the composite convention: negative indicates more anti-prosecutor coverage), 95% bootstrap CI for the mean difference $[0.98, 1.15]$. This medium effect size is the study's largest, exceeding the stance classification effect ($d = -0.341$) and confirming that the evaluative differential operates through the explicit attribution of critical narratives to specific prosecutors.

Per-theme analysis (Figure 10) uses a different metric: the percentage of articles in which *any* of the four attribution methods flags the theme, producing higher prevalence rates than the keyword-proximity measure above. By this article-level flag, recall-related themes appear 4.02 times more often in progressive prosecutor coverage (19.5% vs. 4.8%, $\chi^2 = 641.9$, $p < 10^{-141}$, Cramér's $V = 0.22$). Crime-rising themes appear 1.72 times more often (21.0% vs. 12.2%, $\chi^2 = 178.7$, $p < 10^{-40}$), and soft-on-crime framing is 3.74 times more prevalent ($\chi^2 = 68.5$, $p < 10^{-16}$). The paired county theme comparisons confirm (reported in the standard progressive-minus-traditional

direction, where positive d indicates higher theme attribution for the progressive prosecutor): in San Francisco, Boudin scores higher than Jenkins ($d = 0.28$, $p < .001$), and in Alameda, Price scores higher than O'Malley ($d = 0.58$, $p < .001$), the study's largest within-county effect.

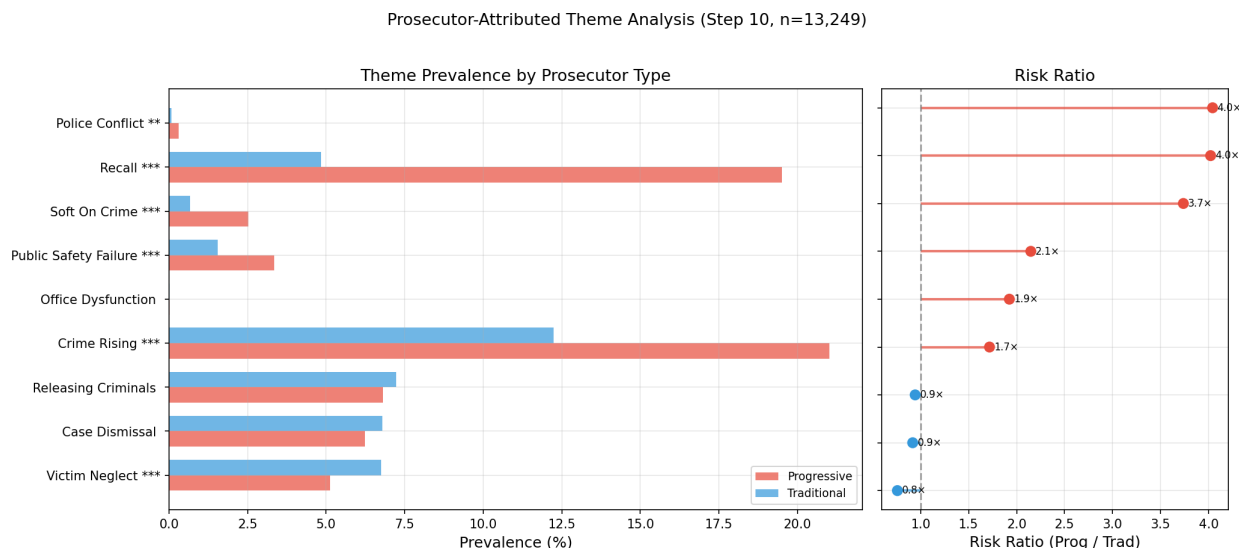


Figure 10: Prosecutor-attributed theme prevalence by ideology (left) and risk ratios (right). Significance: *** $p < .001$, ** $p < .01$, * $p < .05$. Themes sorted by risk ratio.

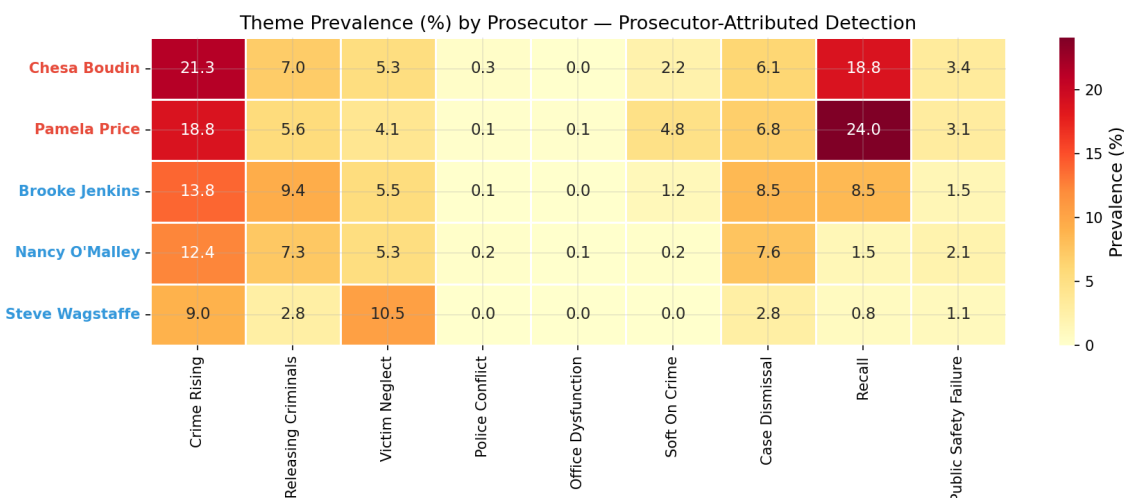


Figure 11: Theme prevalence (%) by individual prosecutor. Rows are colored by ideology (red = progressive, blue = traditional). Darker cells indicate higher prevalence.

Multi-method validation (Figure 12) shows that 41.4% of articles with any detected theme trigger

two or more independent detection methods, confirming convergent evidence rather than single-method artifacts.

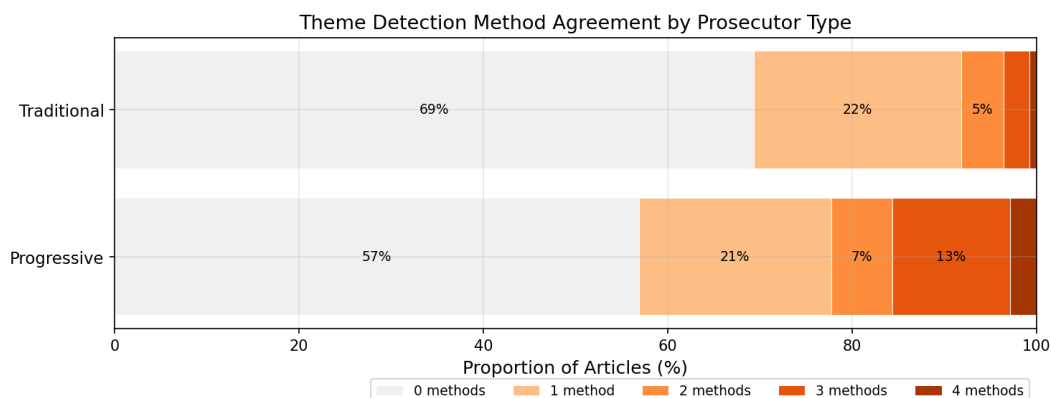


Figure 12: Distribution of detection method agreement by prosecutor type. Segments show the proportion of articles detected by 0–4 independent methods.

4.7 Composite Summary

The preceding per-method analyses (Sections 4.1–4.6) constitute the study’s primary findings and demonstrate that evaluative and tonal dimensions of coverage diverge sharply. The composite below serves as a summary measure for time-series visualization and overall comparison, not as the principal test of differential treatment.

When the four methods are combined into a weighted composite (weights: 0.35 sentiment, 0.30 stance, 0.20 keywords, 0.15 document), the overall group comparison yields $d = -0.157$ ($p < .001$, 95% bootstrap CI $[-0.036, -0.023]$), with a TOST equivalence test confirming the effect falls within conventional equivalence bounds ($p = .007$ at $d = 0.2$; see Section 3.6 for bound justification). Robustness checks across six alternative weighting specifications (Section 3.6) confirm the composite effect is negative and significant under every scheme; the current weights produce one of the most conservative estimates. This composite serves as a proxy for the aggregate reader experience: a reader does not consume “stance” and “sentiment” as separate channels but is exposed to both the emotional tone and the evaluative framing of an article simultaneously. The composite approximates this net impression and provides a single dependent variable for the time series and

overall comparison (Figure 13). However, the per-method results above demonstrate that this aggregate figure obscures the distinct mechanisms at play: it averages a large evaluative effect with a null tonal effect, producing a small overall number that understates the stance differential and overstates the sentiment contribution.

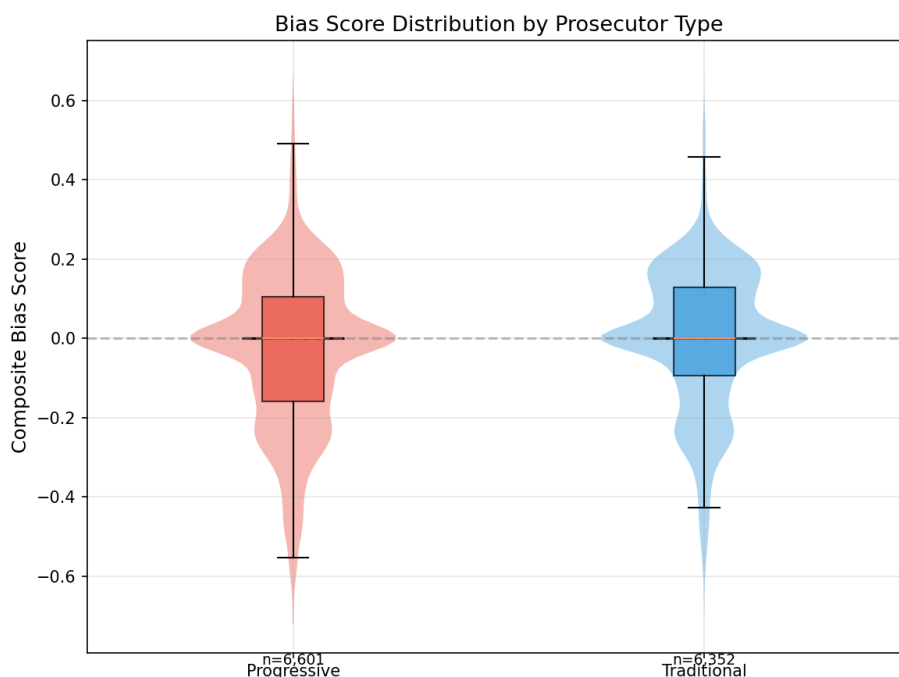


Figure 13: Aggregate tone-evaluation index distributions by prosecutor ideology. Violin plots show the full distribution; inner box plots show the interquartile range. Negative scores indicate more negative coverage.

4.8 Sensitivity to Traditional Baseline Composition

Brooke Jenkins accounts for more than half of the traditional baseline ($n = 3,207$ of 6,352 traditional articles). As documented in Appendix B, Jenkins’s coverage is atypically negative for a traditional prosecutor: her mean composite score (-0.016) sits between the progressive mean (-0.032) and the other traditional prosecutors’ mean ($+0.012$), and she differs significantly from O’Malley and Wagstaffe ($d = -0.15$, $p < 10^{-9}$). This is consistent with the politically charged nature of her appointment as Boudin’s replacement during the recall.

To assess whether this anomalous baseline composition suppresses the between-group effect, I re-computed all per-method comparisons excluding Jenkins. The composite effect increases from $d = -0.157$ to $d = -0.230$ ($p < 10^{-25}$); stance increases from $d = -0.341$ to $d = -0.466$ ($p < 10^{-75}$); keywords from $d = -0.218$ to $d = -0.313$ ($p < 10^{-46}$). The two sentiment measures remain non-significant. The full-sample estimates reported throughout this paper are therefore conservative: the traditional baseline includes a prosecutor whose coverage profile partially resembles that of the progressive group, attenuating the observed differential.

5 Discussion

This study provides the first large-scale, multi-method analysis of media coverage of prosecutors by ideological orientation. Three principal findings emerge.

First, media bias toward prosecutors is multi-dimensional, and the dimensions diverge. The study’s central contribution is demonstrating that what is commonly treated as a unitary construct, “media bias,” decomposes into at least two independent dimensions: evaluative framing and emotional tone. Stance classification detects a medium effect ($d = -0.341$) that survives regression controls ($\beta = -0.132$, $p < .001$); sentiment analysis detects no effect ($d = 0.037$, $p = .058$; $\beta = 0.008$, $p = .618$ after controls). This divergence is not a methodological artifact but a substantive finding: the media covers progressive prosecutors in the same emotional register as their traditional counterparts but evaluates their performance more critically. This pattern aligns with [Entman’s \(1993\)](#) observation that framing operates through selection and salience rather than through overt evaluative language, and with prior scholarship documenting how structural accumulation of story selection, source amplification, and contextual juxtaposition creates critical impressions without lexical negativity ([Beale, 2006](#); [Surette, 2015](#)). The prosecutor-attributed theme analysis provides the most direct evidence of this mechanism: when themes must be explicitly linked to prosecutors through compound linguistic patterns, the effect size increases to $d = -0.43$, confirming that the evaluative differential operates through the *attribution* of critical narratives rather than

through diffuse tonal differences.

The Alameda County comparison illustrates why decomposition matters. The composite paired comparison yields a non-significant $d = -0.048$ ($p = .295$), an apparent null. However, per-method analysis reveals that stance ($d = -0.275$, $p < .001$) and keywords ($d = -0.373$, $p < .001$) detect substantial differential treatment while sentiment pulls in the opposite direction ($d = 0.082$, $d = 0.124$). The composite averages these opposing signals to near-zero. A single-index approach would miss the differential treatment entirely.

Second, evaluative bias is event-driven rather than stable. The per-method temporal analysis (Figure 7) resolves a key ambiguity. Stance classification and keyword analysis show dramatic quarterly variation ($SD = 0.32$ and 0.35 , respectively), with the deepest effects coinciding with the Boudin recall campaign (2021Q4–2022Q2) and the Price recall movement (2023Q2–Q3). Aspect sentiment and document sentiment show less variation ($SD = 0.28$ and 0.22) and no systematic pattern tied to political events. The aggregate temporal heterogeneity (composite range $d = -0.39$ to $+0.84$, $SD = 0.37$) is driven entirely by the evaluative dimensions; tonal bias is absent throughout. This finding suggests that recall campaigns and politically charged events amplify evaluative scrutiny of progressive prosecutors, rather than a stable editorial disposition against them. Accordingly, the term “bias” as used throughout this paper denotes differential evaluative responsiveness to political events, not a stable editorial preference: the finding is not that newsrooms maintain a constant anti-progressive disposition, but that politically charged events trigger evaluative scrutiny of reform prosecutors that has no tonal counterpart in coverage of traditional prosecutors facing comparable circumstances.

Third, framing differences are more pronounced and potentially more consequential than any tonal measure. The Cramér’s V of 0.225 for dominant frame distributions, a medium effect, indicates that progressive and traditional prosecutors are covered through substantively different narrative lenses. Progressive prosecutors are held to an accountability standard (40% of articles vs. 23%) and viewed through an ideological lens (4% vs. 1%), while traditional prosecutors receive

more human-interest framing (36% vs. 21%). These framing choices may matter more than any aggregate tone measure: they determine whether voters receive performance evaluations or human-interest stories about structurally similar officeholders.

5.1 Limitations

Several limitations warrant caution. First, this is an observational study: differences in coverage may reflect genuine differences in prosecutorial conduct, public controversy, or newsworthy events rather than ideological bias per se. Boudin’s recall, Price’s contentious relationship with law enforcement, and the unique political dynamics of Bay Area criminal justice all represent confounds that within-county comparisons can mitigate but not eliminate. Second, the NLP methods employed, while state-of-the-art, have known limitations: zero-shot classifiers may exhibit systematic biases, including the potential confusion of quoted source stance with authorial stance in attribution-heavy journalism (e.g., classifying “critics say Boudin is soft on crime” as the article’s evaluative position rather than a reported viewpoint); sentiment models trained on social media text may transfer imperfectly to news prose, and keyword dictionaries inevitably embed researcher assumptions about what constitutes negative framing. Third, the study examines a single metropolitan area; whether these patterns generalize to other progressive prosecutors (e.g., Krasner in Philadelphia, Foxx in Chicago, Gascón in Los Angeles) is an empirical question. Fourth, news production is highly redundant: a single press conference or recall milestone often generates near-duplicate stories across outlets and wire services, inflating the effective sample size. Although the cluster-robust standard errors (clustered by publication) partially address within-outlet dependence, they do not account for story-level clustering across publications covering the same event. The extreme p -values reported throughout (e.g., $p < 10^{-127}$) should accordingly be interpreted as strong directional evidence rather than precise probability statements; the effect sizes, which are less sensitive to sample size inflation, are the substantively meaningful quantities.

5.2 Implications and Future Directions

These findings have implications for scholarship on prosecutorial politics, media bias, and computational text analysis. For prosecutorial politics, the results suggest that reform prosecutors face a structurally different media environment: one organized around accountability and ideological evaluation rather than the human-interest narratives that characterize coverage of traditional prosecutors. This asymmetry may contribute to the political vulnerability of progressive prosecutors, who face constant performance evaluation in the press while their traditional counterparts benefit from more diffuse, story-driven coverage.

For media bias research, the divergence between sentiment (null) and stance (large effect) highlights the importance of multi-dimensional measurement. Studies relying solely on sentiment analysis may underestimate or miss entirely the evaluative dimensions of media coverage that operate through framing rather than emotional tone.

These findings also raise concerns about democratic accountability in criminal justice. Media coverage is the primary conduit through which the public learns about prosecutorial policy (Wright, 2009), and the results indicate that this conduit applies systematically different evaluative standards depending on a prosecutor’s ideological orientation. When progressive prosecutors are disproportionately covered through accountability frames while traditional prosecutors receive human-interest coverage, voters receive different types of information about structurally similar officeholders, developing calibrated judgments about a reform prosecutor’s policy outcomes while remaining uninformed about a traditional prosecutor’s performance. This asymmetry is compounded by the “toplash” dynamics Goldstein (2024) describes, in which institutional actors leverage media relationships to amplify opposition to reform. If small, persistent shifts in evaluative framing accumulate over thousands of articles, they may contribute to the political vulnerability of reform prosecutors through a mechanism that is invisible in any individual article but consequential in aggregate. Taken together, these results demonstrate that the press covers reform prosecutors differently in kind rather than in degree, not through uniform hostility, but through systematic dif-

ferences in evaluative framing, sourcing, and narrative structure that are invisible to tonal analysis alone.

Future work should expand the corpus to include other reform prosecutors nationally, develop human-coded validation samples to benchmark NLP performance, and pursue causal identification strategies, such as regression discontinuity designs around close prosecutorial elections, to distinguish ideological bias from coverage differences attributable to newsworthy events.

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Appendix A LLM-Based Structural Content Extraction

The multi-method pipeline described in Section 3 measures the *tone* and *framing* of coverage but does not directly examine the structural building blocks of articles: who is quoted, what claims are made, and with what evidence. Understanding these structural features is essential for explaining *why* tone differs across prosecutor types, yet manual content coding at the scale of 12,953 articles is prohibitively expensive. This appendix reports results from a large-scale LLM-based structured extraction that decomposes article content into discrete, categorized elements.

Method

I use Google’s langextract library, which leverages Gemini 2.5 Flash to extract structured information from text with source grounding: every extraction is anchored to a specific span in the original article. The model was prompted with a detailed extraction schema and three diverse few-shot examples (one critical of a progressive prosecutor, one neutral about a traditional prosecutor, one with mixed coverage) to calibrate extraction behavior across ideological contexts. I applied the extraction pipeline to 4,763 prosecutor-attributed articles (3,423 progressive, 1,340 traditional), covering all five prosecutors. The extraction was conducted in batches due to API rate limits; articles were selected in prosecutor-stratified order (processing each prosecutor’s full set sequentially) until API quotas were exhausted, yielding extraction rates of approximately 52% for the two progressive prosecutors (Boudin 2,987/5,773; Price 436/828) and 21% for the three traditional prosecutors (Jenkins 685/3,207; O’Malley 302/1,497; Wagstaffe 353/1,648). Because extraction rates differ across prosecutors, all comparisons below use per-article rates rather than raw counts.

Extraction prompt

The following prompt was passed to the model for every article. It defines five extraction classes, each with structured attributes that constrain the model's output to a fixed schema:

Extract the following types of information from this news article about a district attorney / prosecutor. Use exact text spans from the article.

1. `claim_against_prosecutor`: Specific accusations, criticisms, or negative claims made about the prosecutor's performance, policies, or character.

Attributes: `claim_type` (performance | policy | character | competence), `specificity` (vague | specific | quantified), `evidence_cited` (none | anecdotal | statistical | official_report).

2. `source_attribution`: Identify who is speaking, quoted, or cited as a source.

Attributes: `source_type` (police | victim | defense_attorney | prosecutor | politician | community_member | journalist | expert | advocacy_group), `stance_toward_prosecutor` (critical | supportive | neutral).

3. `causal_claim`: Claims that the prosecutor's actions caused or contributed to some outcome.

Attributes: `effect` (crime_increase | public_safety_decline | case_outcome | community_impact | positive_outcome), `causal_strength` (explicit | implied | speculative), `direction` (prosecutor_caused_harm | prosecutor_helped | ambiguous).

4. `policy_action`: Concrete policies, decisions, or actions attributed to the prosecutor.

Attributes: `action_type` (declined_to_prosecute | reduced_charges | new_policy | fired_staff | reversed_predecessor | enhanced_prosecution | other), `domain` (drugs | property_crime | violent_crime | bail | sentencing | staffing | juvenile | general), `framing` (positive | negative | neutral).

5. `comparison`: Explicit comparisons between the current prosecutor and a predecessor, another prosecutor, or a general standard.

Attributes: compared_to (predecessor | other_prosecutor | general_standard), dimension (toughness | case_outcomes | policy | competence | ideology), who_favored (current | predecessor | neither).

Only extract items that are clearly present. Do not fabricate or infer. Extract text spans in order of appearance. Do not paraphrase.

Few-shot examples

Three examples were provided with each API call to demonstrate the expected extraction format. Each example pairs a synthetic article excerpt with the complete set of extractions the model should produce. The examples were designed to span the ideological and tonal range of the corpus.

Example 1: Critical coverage of a progressive prosecutor.

“San Francisco’s District Attorney Chesa Boudin faced renewed criticism this week after police data showed a 15% increase in car break-ins since he took office. Police Officers Association president Tony Montoya said officers are demoralized because suspects they arrest are quickly released without charges. ‘Why bother making arrests when the DA won’t prosecute?’ Montoya asked. Boudin defended his record, saying his office focuses on serious violent crime rather than low-level offenses. Recall organizers said the crime statistics prove Boudin’s progressive policies have failed San Francisco residents.”

Class	Extracted Text	Key Attributes
claim_against	“a 15% increase in car break-ins since he took office”	performance, quantified, statistical
source_attrib.	“Police Officers Association president Tony Montoya said officers are demoralized”	police, critical
source_attrib.	“Boudin defended his record. . .”	prosecutor, supportive
causal_claim	“the crime statistics prove Boudin’s progressive policies have failed”	crime_increase, explicit, harm
policy_action	“suspects they arrest are quickly released without charges”	declined_to_prosecute, negative
source_attrib.	“Recall organizers said. . .”	advocacy_group, critical

Example 2: Neutral coverage of a traditional prosecutor.

“Brooke Jenkins, who replaced Boudin after the recall, announced a new initiative targeting retail theft rings in the downtown area. Jenkins said her office would file felony charges against organized shoplifting groups, reversing Boudin’s practice of treating most retail theft as misdemeanors. Supervisor Matt Dorsey praised the move. Defense attorney Niki Solis warned that harsher penalties would disproportionately affect low-income communities without reducing crime.”

Class	Extracted Text	Key Attributes
policy_action	“file felony charges against organized shoplifting groups”	enhanced_prosecution, property_crime, positive
comparison	“reversing Boudin’s practice of treating most retail theft as misdemeanors”	predecessor, toughness, current
source_attrib.	“Supervisor Matt Dorsey praised the move”	politician, supportive
source_attrib.	“Defense attorney Niki Solis warned...”	defense_attorney, critical

Example 3: Mixed coverage of a progressive prosecutor.

“Alameda County DA Pamela Price faces a recall effort barely a year into her term. Critics say Price has been too lenient, pointing to cases where violent offenders received reduced sentences. The Save Alameda for Everyone committee cited three homicide cases where Price’s office offered plea deals. Price responded that her office has actually increased the conviction rate for violent felonies compared to her predecessor Nancy O’Malley.”

Class	Extracted Text	Key Attributes
claim_against	“Price has been too lenient”	policy, vague, none
claim_against	“violent offenders received reduced sentences”	performance, specific, anecdotal
source_attrib.	“The Save Alameda for Everyone committee cited three homicide cases”	advocacy_group, critical
policy_action	“Price’s office offered plea deals”	reduced_charges, violent_crime, negative
comparison	“increased the conviction rate... compared to her predecessor”	predecessor, case_outcomes, current
source_attrib.	“Price responded that her office has actually increased...”	prosecutor, supportive

The extraction schema defines five content categories: (1) *claims against the prosecutor*, classified by type (performance, policy, character, competence) and evidence quality; (2) *source attributions*, identifying the speaker’s role and stance toward the prosecutor; (3) *causal claims* linking the prosecutor’s actions to outcomes; (4) *policy actions* attributed to the prosecutor; and (5) *comparisons* between the current prosecutor and predecessors or standards. The pipeline produced 48,642 total extraction instances across the 4,763 articles (mean = 10.2 per article).

Results

Source type distribution. The types of sources quoted differ significantly between prosecutor groups ($\chi^2 = 360.2$, $p < 10^{-72}$, $df = 8$; see Figure 14). Because the extraction sample is imbalanced by design (3,423 progressive vs. 1,340 traditional articles), all comparisons are expressed as proportions of total sources within each group and as per-article rates:

Coverage of progressive prosecutors features proportionally more expert sources ($1.9\times$), advo-

Table 5: Source type distribution by prosecutor ideology. Proportions are within-group; rates are per attributed article (progressive $n = 3,423$; traditional $n = 1,340$).

Source Type	Prog. %	Trad. %	Prog./art.	Trad./art.	Ratio
Police	11.9%	12.6%	0.74	0.66	1.1×
Victim	2.6%	2.2%	0.16	0.12	1.3×
Defense attorney	7.0%	8.5%	0.43	0.44	1.0×
Prosecutor (self)	33.2%	41.9%	2.06	2.18	0.9×
Politician	10.2%	6.8%	0.63	0.35	1.8×
Community member	11.8%	11.7%	0.73	0.61	1.2×
Expert	8.1%	4.9%	0.50	0.26	1.9×
Journalist	6.7%	4.9%	0.41	0.25	1.6×
Advocacy group	6.5%	3.9%	0.40	0.20	2.0×

cacy group voices (2.0×), and politician quotes (1.8× per article). Police appear at roughly equal per-article rates; defense-attorney and self-quote proportions are also similar. These sourcing differentials suggest that progressive prosecutor coverage draws on a broader and more politically diverse source ecology. This is consistent with the accountability and reform framing patterns documented in Section 4, and with the “toplash” dynamics described by Goldstein (2024): the elevated presence of expert, advocacy, and politician sources around progressive prosecutors is precisely the source ecology one would expect when institutional actors mobilize media relationships to contest reform agendas. The distribution of source *stance* further illuminates the sourcing asymmetry:

Table 6: Source stance per article by prosecutor ideology (progressive $n = 3,423$; traditional $n = 1,340$).

Source Stance	Prog./art.	Trad./art.	d	p
Critical	1.64	1.11	0.19	$< 10^{-8}$
Supportive	1.35	0.78	0.27	$< 10^{-15}$
Neutral	3.20	3.31	-0.03	.300

Progressive prosecutors attract both more critical and more supportive sources per article, while neutral sources are equally represented, reflecting the greater political salience and controversy surrounding progressive prosecutors rather than a one-sided sourcing imbalance.

Causal claims. Progressive prosecutors face 2.3 times as many “prosecutor caused harm” attributions per article (0.43 vs. 0.19). “Prosecutor helped” attributions are also 1.5 times higher per

article for progressives (0.14 vs. 0.09). The asymmetric harm ratio provides a specific structural mechanism through which progressive prosecutors receive more negative coverage.

Claim types. Per article, progressive prosecutors face more claims across all categories: performance (0.45 vs. 0.29, 1.6 $\times$), policy (0.48 vs. 0.28, 1.7 $\times$), character (0.42 vs. 0.35, 1.2 $\times$), and competence (0.31 vs. 0.25, 1.2 $\times$). The largest differentials appear for performance and policy criticisms, suggesting that progressive prosecutors are disproportionately criticized for *what they do*, their decisions and policy choices, rather than for personal attributes, a pattern consistent with the accountability framing differential observed in Section 4.

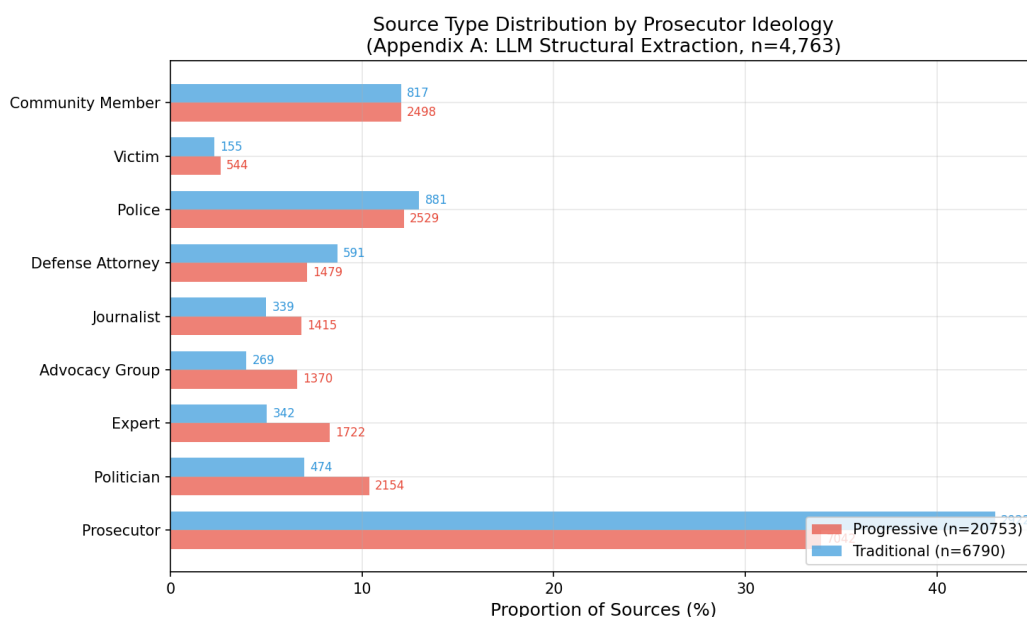


Figure 14: Source type distribution by prosecutor ideology from LLM structural extraction ($n = 4,763$ articles). Bars show proportions of total sources within each group; annotations show raw counts. The largest relative differentials are in expert sources (1.9 $\times$), advocacy groups (2.0 $\times$), and politicians (1.8 $\times$ per article).

Per-prosecutor extraction density. Pamela Price generates the most extraction-dense coverage (12.4 per article, $n = 436$), followed by Chesa Boudin (10.7, $n = 2,987$), Nancy O'Malley (9.1, $n = 302$), Brooke Jenkins (8.9, $n = 685$), and Steve Wagstaffe (7.1, $n = 353$).

Limitations

The extraction sample is imbalanced (3,423 progressive vs. 1,340 traditional articles), reflecting the composition of the underlying corpus; all comparisons are expressed as per-article rates or proportions. Because articles were processed in each prosecutor’s corpus order rather than drawn as a stratified random sample, the extracted subset may not be fully representative along temporal or event dimensions; a stratified re-extraction (by prosecutor, quarter, and outlet) would provide stronger assurance that the sourcing and claim-structure patterns generalize to the full corpus. The source type asymmetry, particularly the expert and advocacy-group disparities, suggests that tone differences detected in the main analysis may partly reflect differential journalistic sourcing practices rather than editorial bias per se.

Appendix B LLM-Based Bias Indicator Extraction (Pilot Study)

The structural extraction in Appendix A characterizes *what* articles contain; this appendix reports a complementary pilot that asks *how* the coverage is biased. A well-sourced, evidence-rich article can be legitimately critical of a prosecutor without being biased, while a superficially neutral article can exhibit bias through ungrounded assertions, source imbalance, loaded language, or omission of relevant context.

Method

Using the same langextract + Gemini 2.5 Flash infrastructure, I designed a schema targeting four categories of bias indicator: (1) *ungrounded negative claims*; (2) *source prominence imbalance*; (3) *loaded language*; and (4) *missing context*. A key methodological challenge is preventing the model from conflating legitimate criticism with bias, addressed through three calibration examples demonstrating that criticism with evidence is not bias. The extraction was applied to a balanced stratified sample of 200 articles (100 progressive, 100 traditional).

Extraction prompt

The following prompt was passed to the model for every article. Unlike the structural extraction in Appendix A, this schema explicitly distinguishes legitimate criticism from bias through calibration guidance:

Analyze this news article about a district attorney / prosecutor for BIAS INDICATORS -- patterns of coverage that reveal systematic slant rather than fair reporting. Use exact text spans from the article.

Only extract genuine bias indicators. Legitimate critical reporting with evidence is NOT bias. Balanced articles may have zero extractions.

1. `ungrounded_negative_claim`: A negative assertion about the prosecutor that lacks adequate supporting evidence within the article.

Attributes: `claim_content` (performance | policy | character | competence), `evidence_quality` (none | anonymous_source | single_anecdote | stats_without_baseline | adequately_sourced), `systemic_blame` (true | false).

2. `source_prominence_imbalance`: A source given disproportionate prominence relative to other viewpoints, creating an imbalanced picture.

Attributes: `source_stance` (critical | supportive), `prominence_mechanism` (placement_lede | placement_closing | extended_quote | sole_named_source | no_counterbalance | headline_framing), `counterbalance_present` (true | false).

3. `loaded_language`: Emotionally charged, non-neutral word choices that go beyond factual reporting.

Attributes: `language_type` (pejorative_label | scare_quotes | presuppositional_verb | hyperbole | ideological_framing | dehumanizing), `target` (prosecutor | prosecutor_policy | prosecutor_supporters | other), `position` (headline_or_lede | body | closing).

4. `missing_context`: Important context that is conspicuously absent, making the coverage

misleading.

Attributes: what_is_missing (trend_context | comparison_baseline | policy_rationale | systemic_factors | legal_constraints | prosecutor_response | alternative_explanation), claim_it_supports (free text).

CALIBRATION GUIDANCE:

- A well-sourced article that is critical of a prosecutor is NOT biased if the criticism is grounded in evidence and the prosecutor's perspective is included.
- An article that is favorable to a prosecutor IS biased if it cherry-picks achievements, omits failures, or relies on uncritical source selection.
- Apply the same standard regardless of which prosecutor is covered.

Only extract items that are clearly present. Do not fabricate or infer. Extract text spans in order of appearance. Do not paraphrase.

Few-shot examples

Three examples were provided to calibrate the model's threshold for flagging bias. Critically, Example 2 demonstrates that a well-sourced critical article should produce *minimal* bias indicators, preventing the model from conflating criticism with bias.

Example 1: Heavily biased anti-progressive article.

"San Francisco's failed experiment: How Chesa Boudin let criminals run wild. Since the radical progressive prosecutor took office, residents say the city has become unrecognizable. Car break-ins are rampant, shoplifters brazenly clear shelves with impunity, and violent offenders walk free hours after arrest. 'He cares more about criminals than victims,' said retired officer Jim Walsh, who spent 30 years on the force. Walsh described case after case of suspects released without charges. A business owner on Market Street, who asked not to be named for

fear of retaliation, said Boudin’s policies have destroyed the neighborhood. Critics point to a 12% rise in property crime, though experts note similar trends across major cities regardless of prosecutor ideology. Boudin’s office did not respond to requests for comment.”

Class	Extracted Text	Key Attributes
loaded_lang.	“San Francisco’s failed experiment”	presuppositional_verb, policy, headline
loaded_lang.	“let criminals run wild”	hyperbole, prosecutor, headline
loaded_lang.	“the radical progressive prosecutor”	pejorative_label, prosecutor, body
ungrounded	“shoplifters brazenly clear shelves with impunity, and violent offenders walk free hours after arrest”	performance, none, systemic_blame
source_imbal.	“retired officer Jim Walsh...case after case of suspects released”	critical, extended_quote, no counterbalance
ungrounded	“A business owner...asked not to be named...Boudin’s policies have destroyed the neighborhood”	policy, anonymous_source
missing_ctx.	“Critics point to a 12% rise in property crime”	comparison_baseline
missing_ctx.	“Boudin’s office did not respond to requests for comment”	prosecutor_response

Example 2: Well-constructed critical article (minimal bias).

“Alameda County DA Pamela Price’s first year in office has drawn sharp scrutiny. An East Bay Times analysis of court records found that Price’s office offered plea deals in 73% of violent felony cases, compared to 58% under predecessor Nancy O’Malley. The analysis examined 412 cases filed between January and September 2023. Victims’ rights advocate Maria Chen

said the plea rates concern families who expected tougher prosecution. Price defended the approach in a press conference, saying her office prioritizes cases with the strongest evidence and that conviction rates on cases taken to trial have actually increased to 89%. Criminal justice professor David Lang at UC Berkeley noted that higher plea rates do not necessarily indicate leniency, as they may reflect more realistic case assessment. The recall campaign against Price has seized on the plea statistics.”

Class	Extracted Text	Key Attributes
missing_ctx.	“The recall campaign against Price has seized on the plea statistics”	systemic_factors (plea rates attributable to case composition, not ideology alone)

This example is critical for calibration: the article contains sharp criticism grounded in court records, includes the prosecutor’s response and an expert perspective, and produces only one extraction (a minor missing-context flag). This teaches the model that evidence-based critical reporting is *not* bias.

Example 3: Pro-traditional puff piece.

“New DA Brooke Jenkins is restoring order to San Francisco. In her first six months, the tough-on-crime prosecutor has brought a sense of accountability that residents desperately needed. Jenkins announced felony charges against a prolific shoplifter, drawing praise from the business community. ‘Finally, someone who takes our concerns seriously,’ said Union Square Alliance director Tom Richards, who described Jenkins as a breath of fresh air after the chaos of the Boudin era. Police Chief Bill Scott lauded the improved cooperation between his department and the DA’s office. Crime data for the period is not yet available from official sources.”

Class	Extracted Text	Key Attributes
loaded_lang.	“restoring order to San Francisco”	presuppositional_verb, prosecutor, headline
loaded_lang.	“the tough-on-crime prosecutor has brought...accountability that residents desperately needed”	ideological_framing, prosecutor, body
source_imbal.	“Tom Richards... a breath of fresh air after the chaos of the Boudin era”	supportive, extended_quote, no counterbalance
source_imbal.	“Police Chief Bill Scott lauded the improved cooperation”	supportive, no_counterbalance
missing_ctx.	“Crime data for the period is not yet available”	trend_context (effectiveness claimed without evidence)

This example demonstrates that the schema applies symmetrically: a favorable article about a traditional prosecutor triggers loaded-language, source-imbalance, and missing-context indicators just as readily as a negative article about a progressive prosecutor.

Results

Primary test. Progressive prosecutors receive a slightly more negative mean bias score ($M = -0.143$) than traditional prosecutors ($M = -0.111$), but the difference is not statistically significant: $t(198) = -0.83$, $p = .406$, $d = -0.118$, bootstrap 95% CI $[-0.105, +0.043]$. The null primary result contrasts with the significant $d = -0.157$ from the main pipeline (Section 4).

Per-indicator decomposition. Individual bias indicators show small, mixed effects (see Figure 15):

Loaded negative language is the only indicator favoring the progressive bias hypothesis ($d = +0.14$): progressive prosecutors receive more emotionally charged negative word choices. Para-

Table 7: Bias indicator means by prosecutor ideology (Appendix B pilot, $n = 200$).

Indicator	Progressive M	Traditional M	d
Ungrounded claims	0.50	0.74	-0.12
Ungrounded (severe)	0.36	0.62	-0.14
Systemic blame	0.23	0.23	0.00
Source imbalance (critical)	0.64	0.67	-0.02
Source imbalance (supportive)	0.10	0.16	-0.16
Loaded language (total)	1.45	1.19	+0.10
Loaded language (negative)	0.55	0.38	+0.14
Loaded language (headline)	0.24	0.25	-0.02
Missing context	0.77	0.95	-0.12

doxically, ungrounded claims are *higher* for traditional prosecutors ($d = -0.12$; the severe subcategory specifically: $d = -0.14$). The distribution of *prominence mechanisms* differs significantly across groups ($\chi^2 = 14.58$, $p = .012$, $df = 5$): traditional prosecutor coverage relies more on sole named sources (25 vs. 7 instances), while progressive prosecutor coverage exhibits more “no counterbalance” patterns (16 vs. 10).

Language type decomposition. Disaggregating loaded language by type reveals a qualitative distinction in *how* bias manifests. Progressive prosecutor coverage exhibits more presuppositional verbs (34 vs. 24 instances) and ideological framing (41 vs. 24 instances): language that embeds assumptions about the prosecutor’s politics or takes contested claims as given. Traditional prosecutor coverage shows more pejorative labels (35 vs. 27 instances). The *mechanism* of bias thus differs: progressive prosecutors are framed through an ideological lens where their reform identity is constantly invoked and assumed; bias against traditional prosecutors, when it occurs, is expressed more bluntly. Evidence quality further distinguishes the groups: articles about traditional prosecutors more often contain ungrounded negative claims with *no* supporting evidence at all (53 vs. 29 instances).

Per-prosecutor decomposition. The most striking pattern in the pilot data is that Brooke Jenkins, the traditional San Francisco prosecutor, receives the worst-quality coverage in the sample by standard journalistic metrics: 0.93 ungrounded claims per article (vs. Boudin’s 0.47), 0.84 severe

ungrounded claims (vs. 0.35), and 1.12 missing context instances (vs. 0.74). By contrast, Nancy O'Malley receives the best-quality coverage of any prosecutor (0.26 ungrounded claims/article, 1.6 total indicators/article). This intra-traditional heterogeneity means that the aggregate null result for the traditional group conceals substantial within-group variation. The Jenkins finding is consistent with the politically charged nature of her appointment: she replaced a highly controversial progressive predecessor and was herself a focal point of the Boudin recall narrative, rather than evidence of systematic pro-progressive favoritism.

Convergent validity. The bias score correlates moderately with the aggregate tone-evaluation index from the main pipeline: Pearson $r = 0.257$, $p = .0002$; Spearman $\rho = 0.237$, $p = .0007$. The two approaches capture overlapping but distinct constructs: the NLP pipeline measures aggregate tone and evaluative stance, while the LLM extraction targets specific journalistic violations.

Per-prosecutor ranking. Pamela Price receives the most negative bias score (-0.158), followed by Chesa Boudin (-0.140), Brooke Jenkins (-0.118), Steve Wagstaffe (-0.104), and Nancy O'Malley (-0.100).

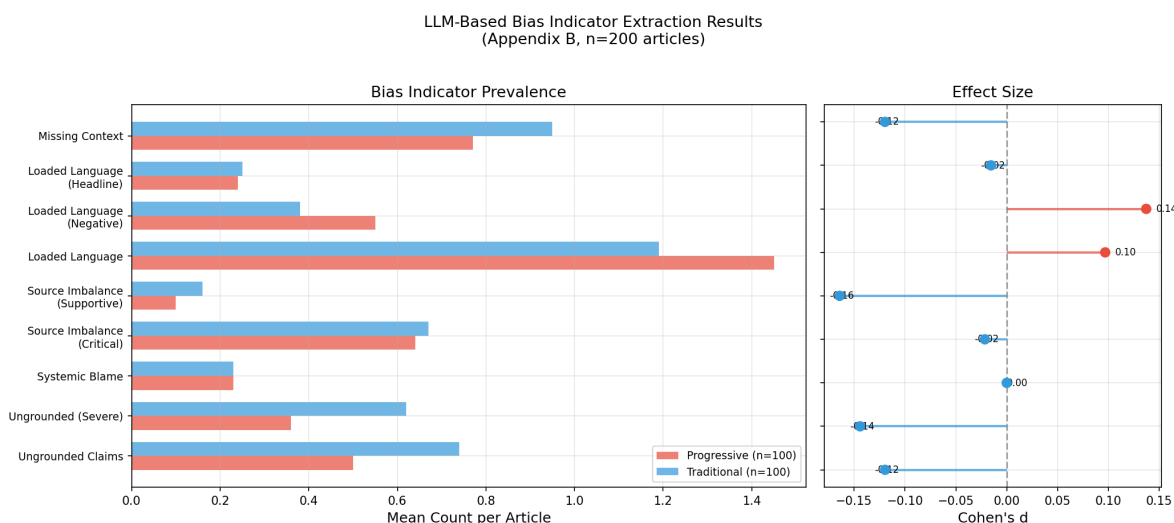


Figure 15: LLM-based bias indicator extraction results (Appendix B pilot, $n = 200$). Left panel: mean indicator counts by ideology. Right panel: Cohen's d effect sizes for each indicator. The only indicator favoring the progressive bias hypothesis is loaded negative language ($d = +0.14$); ungrounded claims are paradoxically higher for traditional prosecutors.

Discussion

The null primary result ($d = -0.118$, $p = .406$) contrasts with the significant $d = -0.157$ from the main pipeline (Section 4). The interpretation is that media bias toward progressive prosecutors operates through tone and framing, the aggregate evaluative register of coverage, rather than through discrete, identifiable journalistic violations (ungrounded claims, source imbalance, missing context).

The loaded language finding ($d = +0.14$) is particularly important here: it provides the mechanistic bridge between what the stance classifier detects as a “vibe” and what is observable in the text as a specific lexical practice. The main pipeline’s stance effect operates through how articles frame prosecutors evaluatively; the loaded language result identifies the textual mechanism: negative word choices carrying evaluative loading beyond their denotative content, presuppositional verbs that embed assumptions about performance, and ideological labels that invoke the prosecutor’s reform identity as a frame for criticism. Progressive prosecutors are not simply described more harshly; they are described through language that *presupposes* rather than asserts the critical evaluation. This is exactly the kind of subtle framing effect that aggregate tone measures (ungrounded claims, missing context) would not capture, because the individual sentences may be technically accurate. Loaded language is therefore not a minor corroborating finding but the observable textual trace of the stance mechanism: the mechanism by which evaluative framing enters coverage without constituting a discrete journalistic violation. The moderate convergent validity ($r = .26$) between the two pipelines is consistent with this view: they overlap because loaded language is tone-adjacent, and they diverge because the rest of the bias indicator schema targets a different domain.

The per-prosecutor and language-type decompositions further reinforce this. Progressive prosecutor bias operates through ideological framing and presuppositional language, subtle mechanisms that do not register as discrete violations. Traditional prosecutors, particularly during politically contested transitions (Jenkins replacing Boudin), are not immune from low-quality coverage; the

aggregate null result conceals substantial within-group heterogeneity. These findings reinforce the main paper’s argument that bias detection requires multi-dimensional measurement: an approach sensitive only to explicit violations would miss the evaluative framing that drives the main effects.

This remains a pilot study ($n = 200$). The Alameda comparison lacks statistical power ($n = 15$ and $n = 23$). A full-corpus extraction is planned to achieve adequate power and enable publication-level subgroup analyses.

Appendix C Illustrative Examples of Identified Bias by Method

This appendix presents concrete examples from the corpus to illustrate what each method detects. Examples are drawn from articles at the extremes of each method’s score distribution and are intended to make the abstract measurement pipeline tangible.

Method A: Aspect-Based Sentiment Analysis

Method A extracts three-sentence context windows around prosecutor mentions. The following window scored $A = -0.932$ (*San Francisco Chronicle*, about Pamela Price):

“Don’t blame Alameda DA Price for crime. The pandemic changed everything, and we failed low-opportunity communities, especially our youth; crime in our cities has climbed in some areas and it is frightening.”

Although the article’s argument is *defensive* of Price, the RoBERTa model responds to the emotional valence of surrounding language, including “crime,” “climbed,” and “frightening,” rather than the argumentative intent. By contrast, a reader letter produced $A = +0.955$: “I am so thankful for Alameda County District Attorney Pamela Price. She has taken on an enormous job and she is doing it well.”

Method B: Zero-Shot Stance Classification

A paragraph from kron4.com received a critical stance score of $B = -0.996$:

“[The letter] cites failed leadership, the movement to defund the police and the district attorney’s unwillingness to charge and prosecute people who commit life threatening crimes.”

BART-MNLI classifies this as criticizing the prosecutor’s handling of crime with near-certainty. The divergence between Methods A and B ($d = 0.037$ for sentiment vs. $d = -0.341$ for stance) is visible in individual examples: an article can score near zero on sentiment while scoring highly critical on stance.

Method C: Enhanced Keyword Analysis

An article about Nancy O’Malley (berkeleyside.org) triggered three themes simultaneously: **soft-on-crime** (“very **light sentences** or no sentences at all”), **releasing criminals** (“Boudin has **re-leased** a person accused of a crime”), and **recall** (“announced a **recall campaign** against O’Malley”). The keywords capture thematic framing rather than sentiment: “light sentences” is emotionally neutral but thematically codes the soft-on-crime narrative.

Method D: Document-Level Sentiment

Because crime reporting is inherently negative, Method D reveals a paradoxical pattern: articles about traditional prosecutors are slightly *more* negative in overall tone ($d = +0.047$, $p = .008$). This underscores why prosecutor-specific methods are necessary.

Media Framing Analysis

The framing analysis classifies paragraphs (including standalone headlines) into five narrative frames. Illustrative examples include:

Table 8: Illustrative examples of each dominant media frame (Appendix C).

Frame	Example Headline	Prosecutor	Score
Accountability	“SF DA Boudin Blamed by Some for Release of Parolee...”	Boudin	1.00
Conflict	“Jenkins blames Boudin in decision to drop case...”	Jenkins	1.00
Human interest	“Family of woman killed in S.F. New Year’s hit and run...”	Boudin	1.00
Reform	“As new D.A., Jenkins vows to be tougher than Boudin”	Jenkins	1.00
Consequences	“Price’s office lags on providing charging data”	Price	0.99

Progressive prosecutors are nearly twice as likely to receive accountability framing (39.7% vs. 22.9%) and more than three times more likely to receive reform framing (4.3% vs. 1.3%), while traditional prosecutors receive substantially more human-interest coverage (35.6% vs. 21.4%; see also Figure 6).

Prosecutor-Attributed Theme Detection

The theme attribution method ($d = -0.43$) requires anti-prosecutor themes to be explicitly linked to a named prosecutor through compound linguistic patterns, distinguishing it from the proximity-based keyword method (Method C). An article titled “Boudin Blunders: SF DA’s downfall leading up to recall” (*kron4.com*, 2022-06-08) received the corpus’s highest theme attribution score (21.3), with four independent detection methods identifying themes including crime-rising, soft-on-crime, case-dismissal, recall, and public-safety-failure. By contrast, a routine crime article about a man accused of hitting a woman with a car (*San Mateo Daily Journal*, 2021-09-24) attributed to Wagstaffe received a theme score of 0.0, illustrating the method’s specificity in not flagging ordinary crime coverage merely because a prosecutor is mentioned.

LLM-Based Bias Indicator Detection

A *San Francisco Chronicle* article about Tenderloin drug enforcement produced 19 bias indicators, including **loaded language** (“drug pushers” as pejorative label; “a stunning 70% increase” as hyperbole), **ungrounded claims** (“arrested and released with few real consequences” without case data), and **missing context** (“Boudin and Public Defender Mano Raju declined to comment”).

Critically, many articles produce zero indicators: evidence-based critical reporting is not flagged as biased. The method also detects bias *favoring* prosecutors, supporting the finding that bias is not unidirectional.